



BOP Accumulator Nitrogen Precharge

West Tellus — TRAINING-01 NOT ISSUED

TRAINING-01 NOT ISSUED • WD 7,000 ft • Subsea 38°F • SET PRECHARGE 3,725 PSIG @ 78°F (DCB table) • req shear 2,600 • 2026-09-29

Precharge Summary

Accumulator	Type	Size gal	Qty	Method	Subsea precharge (PSIG)	BAR
Blue POD						
Pilot	Piston	3	3	B	4,887	337.0
Surface Manifold Reg. Pilot	Piston	1	1	Surf	surface 850	
Subsea Manifold Reg. Pilot	Piston	1	1	A	3,965	273.4
Surface LMRP Connector Reg. Pilot	Piston	1	1	Surf	surface 400	
Subsea LMRP Connector Reg. Pilot	Piston	1	1	A	3,515	242.4
Surface Stack Connector Reg. Pilot	Piston	1	1	Surf	surface 400	
Subsea Stack Connector Reg. Pilot	Piston	1	1	A	3,515	242.4
Surface Upper Annular Reg. Pilot	Piston	1	1	Surf	surface 400	
Subsea Upper Annular Reg. Pilot	Piston	1	1	A	3,515	242.4
Surface Lower Annular Reg. Pilot	Piston	1	1	Surf	surface 400	
Subsea Lower Annular Reg. Pilot	Piston	1	1	A	3,515	242.4
Yellow POD						
Pilot	Piston	2.5	3	B	4,887	337.0
Surface Manifold Reg. Pilot	Piston	0.5	1	Surf	surface 850	
Subsea Manifold Reg. Pilot	Piston	0.5	1	A	3,965	273.4
Surface LMRP Connector Reg. Pilot	Piston	0.5	1	Surf	surface 400	
Subsea LMRP Connector Reg. Pilot	Piston	0.5	1	A	3,515	242.4
Surface Stack Connector Reg. Pilot	Piston	0.5	1	Surf	surface 400	
Subsea Stack Connector Reg. Pilot	Piston	0.5	1	A	3,515	242.4
Surface Upper Annular Reg. Pilot	Piston	0.5	1	Surf	surface 400	
Subsea Upper Annular Reg. Pilot	Piston	0.5	1	A	3,515	242.4
Surface Lower Annular Reg. Pilot	Piston	0.5	1	Surf	surface 400	
Subsea Lower Annular Reg. Pilot	Piston	0.5	1	A	3,515	242.4
LMRP						
Upper Annular Surge	Piston	15	1	A	3,515	242.4
Lower Annular Surge	Piston	15	1	A	3,515	242.4
Lower Stack						
Inner Choke/Kill FSV	Piston	15	1	C	5,293	364.9
Outer Choke/Kill FSV	Piston	15	1	C	5,293	364.9
Dedicated Shear	DCB	28	4	C/OEM	3,725	256.8
Inner Choke/Kill FSV	Piston	15	1	C	5,293	364.9
Outer Choke/Kill FSV	Piston	15	1	C	5,293	364.9
Acoustic Pod Pilot	Piston	2.5	2	B	4,547	313.5
Acoustic NON-Shear	Piston	160	2	B	4,250	293.0
Acoustic NON-Shear	Piston	80	1	B	4,250	293.0
WHC / Wellhead Connector Surge	Piston	15	1	A	3,515	242.4
EHBS Timing Bottle	Piston	2.5	1	OEM	PIB (non-API)	
EHBS Pilot	Piston	0.5	1	B	4,887	337.0

Dedicated Shear - DCB Precharge & Method C Check

SET PRECHARGE (DCB TABLE): 3,725 PSIG (gauge - charge to this) / 3,740 PSIA @ 78°F - CHECKS PASS

Read from the OEM DCB table at the surface set temp - charge to 3,725 PSIG (table value 3,704, rounded up to the nearest 25 psi for gauge charging). EHBS fires a single shear ram (rig has other shear rams, not used in the EHBS sequence) - FVR = one ram, no timing. After Method C adiabatic drawdown, the shear ram closes at 4,045 > required shear 2,600; usable >= FVR 50.60 gal.

FVR function	Gal x DF	Total
EHBS shear ram (EHBS fires one shear - no timing)	46.00 x 1.1	50.60
Total FVR required (single shear ram)		50.60 gal

Method C drawdown (adiabatic) at the DCB-table precharge	Pressure	Note
Condition 0 - surface precharge (table)	3,725 PSIG	@ 78°F, OEM table
Subsea precharge P1 - NOV AX080263 field procedure	3,333 psia	required precharge @ depth (subsea 38°F); pair P1 only with this T1
Condition 1 - charged subsea	4,945 PSIG	supply + control-fluid head
Condition 2 - Shear ram close (after FVR)	4,045 PSIG	adiabatic end

Check	Value	Criterion	Margin	Result
Set precharge (DCB table @ surf temp)	3,725 PSIG	table lookup	-	-
Shear ram closing (Condition 2) - remaining hyd. pressure after shear	4,045 PSIG	> required shear 2,600	+1,445 PSIG	PASS
Usable fluid above required shear	199.29 gal calc. 111.30 gal & cap	>= FVR 50.60 gal	+60.7 gal	PASS

NOV Contractual Petrobras Checks - acoustic two shears, Method B

NA - the CSR does not seal

NOV CONTRACTUAL PETROBRAS CHECK - PASS Rev 05: "Petrobras Spec Requirement: Acoustic Two Shears, accumulator calculations use Method B" - isothermal at 38°F, on the DCBs and the acoustic piston bank combined. DCB 3,725 PSIG · acoustic piston 4,250 PSIG · hydraulic charged 4,945 PSIG · DCB charge volume-limited at the mechanical stop.

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Both cuts must meet the minimum shear pressure - that is what governs, per Rev 05's sequence block ("Pressure Required 2,933 psig" is the SHEAR figure). The seal row is informational and does not govern: Rev 05 takes Tellus's low-pressure-seal minimum from an ER table (ER01593 / 16582494-DAS, 440 psig at zero wellbore rising to 2,227 at 8,354 psia), and the MOPFLPS + MAWHP/6.46 formula shown here under-reads against it. dcbSeq stays false: the sequenced path would let the

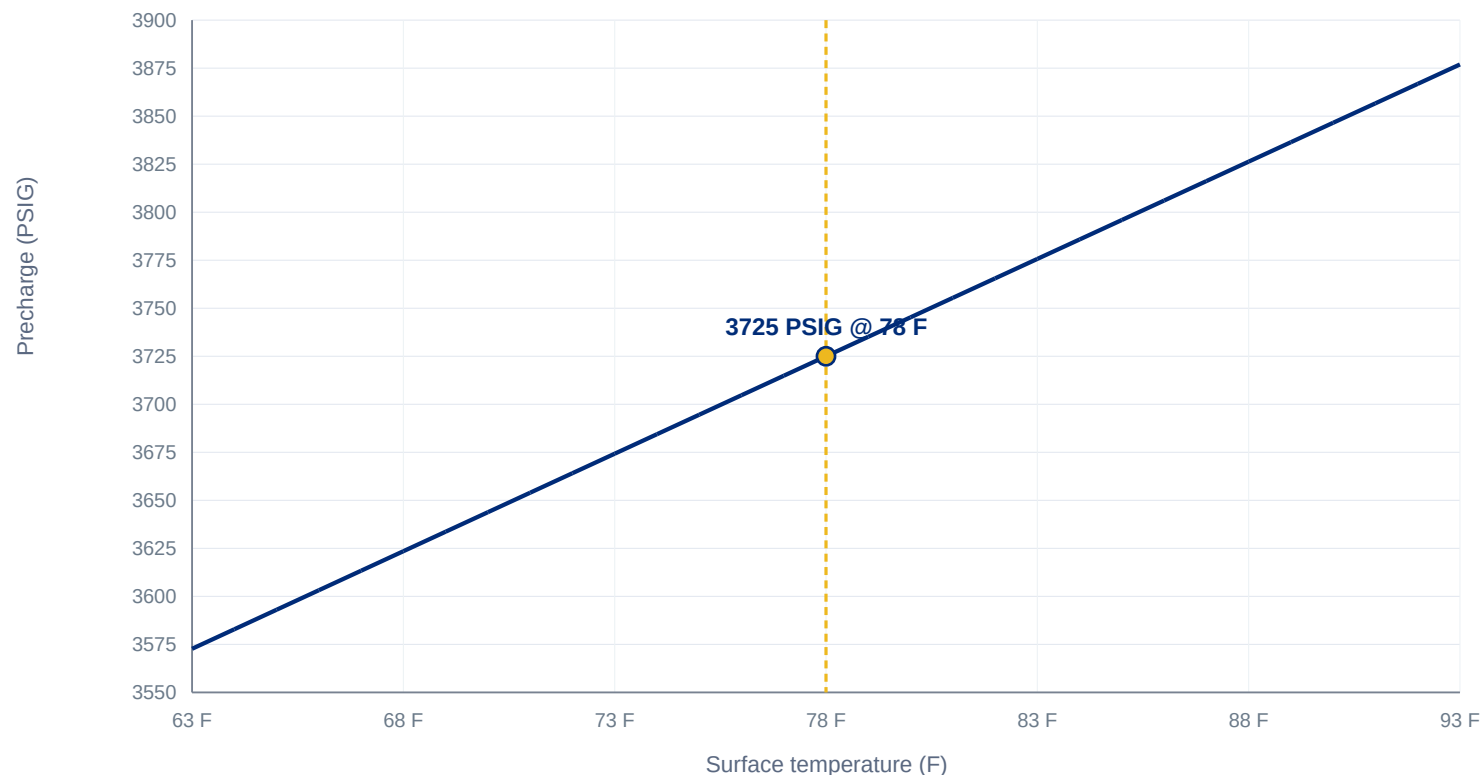
Case	FVR	FVR + 10%	P required	P remaining	Usable: piston + DCB	Result
CSR alonebinding case in Rev 05	46.00 gal	50.60 gal	2,659 PSIG	4,480 PSIG+1,821	49.87 gal + 111.30 gal = 161.18 gal+110.58 gal	PASS
CSR + USR	92.00 gal	101.20 gal	2,659 PSIG	4,119 PSIG+1,460	49.87 gal + 111.30 gal = 161.18 gal+59.98 gal	PASS
CSR + LSR	92.00 gal	101.20 gal	2,659 PSIG	4,119 PSIG+1,460	49.87 gal + 111.30 gal = 161.18 gal+59.98 gal	PASS
Completion (CSR + USR/LSR + stabs + LMRP unlatch)LMRP connector FUNCTIONS remain - it is the BANK that is not fitted	104.12 gal	114.53 gal	3,000 PSIG	4,033 PSIG+1,033	39.85 gal + 111.30 gal = 151.16 gal+36.63 gal	PASS

Acoustic two-shear sequence (Petrobras) - cumulative drawdown	Vol	Cumulative	Pressure at step	Min shear req	Seal (informational)	Result
1 - ShearCasing Shear Ram (CSR), closing ratio 6.46	50.60 gal	50.60 gal	4,480 PSIG	2,659 PSIG	NA - the CSR does not seal	PASS
Timingtiming / delay bottle fill	2.75 gal	53.35 gal	4,459 PSIG	-	-	-
2 - ShearBlind Shear Ram (BSR), closing ratio 6.46	50.60 gal	103.95 gal	4,101 PSIG	2,659 PSIG	1,044 PSIG - seals	PASS

Precharge vs Temperature - +/-15F Operating Window

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The accumulator holds a fixed gas charge. As the surface temperature is subject to change during the precharge process, target the precharge for the ACTUAL temperature at that moment - read off the curve / table below. Calculated set point marked.



F	C	Precharge PSIG	BAR
63	17	3573	246.3
65	18	3593	247.7
67	19	3613	249.1
69	21	3634	250.5
71	22	3654	251.9
73	23	3674	253.3
75	24	3695	254.7
77	25	3715	256.1
79	26	3735	257.5
81	27	3755	258.9
83	28	3776	260.3
85	29	3796	261.7
87	31	3816	263.1
89	32	3836	264.5
91	33	3857	265.9
93	34	3877	267.3

Capacity margin & shear headroom

Precharge regime: OPTIMUM - precharge below bottle limit

Usable (governing bank) 111.3 gal vs FVR 50.6 gal -> margin +60.7 gal (120%)

Max shear deliverable at this precharge ~ 4,025 PSIG (required 2,600 -> headroom +1,425 PSIG)

Precharge still tracks required shear; the margin thins once it reaches the RWP cap.